

Deep Learning for Automatic Modulation Classification

Comparing CNN, LSTM, and Hybrid Architectures on RadioML 2016.10a

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Project Overview

This project implements and compares three deep learning architectures for automatic modulation classification (AMC) — a key capability for cognitive radio, spectrum sensing, and intelligent wireless networks. Using the RadioML 2016.10a dataset (220,000 samples, 11 modulation types, SNR range -20 to +18 dB), I evaluated how well different neural networks can identify modulation schemes directly from raw I/Q signal data.

Key Results

Model	Overall	High SNR (≥ 0 dB)	Low SNR (< 0 dB)
1D-CNN	49.8%	72.7%	26.6%
LSTM	11.1%	13.0%	9.2%
Hybrid CNN-LSTM	21.1%	24.8%	17.5%

Classification accuracy across SNR conditions

Main finding: The 1D-CNN significantly outperforms both LSTM and hybrid architectures. At high SNR conditions, the CNN achieves 72.7% accuracy — approaching practical usability. The pure LSTM fails entirely (11.1% $\approx$ random guessing), while the hybrid shows that CNN feature extraction helps, but adding LSTM layers provides no additional benefit.

Technical Approach

- **1D-CNN:** Two convolutional layers (64 filters, kernel size 8), max pooling, dropout regularization, fully connected classifier. Processes I/Q as 2-channel 1D signal.
- **LSTM:** Single layer with 64 hidden units, treating 128 I/Q samples as a time sequence. Designed to test whether temporal dependencies aid classification.
- **Hybrid:** CNN feature extraction followed by LSTM sequence modeling. Tests whether combining spatial and temporal processing improves results.

All models implemented in PyTorch, trained with Adam optimizer ($\text{lr}=0.001$) and cross-entropy loss. Evaluation on held-out test set with stratified train/val/test split.

Key Insights

1. **CNNs excel at I/Q feature extraction.** Convolutional filters effectively learn modulation-specific patterns (constellation geometry, phase transitions) without manual feature engineering. This aligns with prior work showing CNNs' success on signal data.
2. **LSTMs struggle with raw I/Q.** The sequential structure that LSTMs are designed to capture does not appear to be the primary discriminative factor in modulation classification. Raw I/Q samples may lack the

long-range dependencies that benefit from recurrent modeling.

3. SNR threshold matters. Below -10 dB, all architectures perform near random chance. Reliable classification requires $\text{SNR} \geq -6$ dB, with strong performance above 0 dB.

4. QAM confusion is expected. QAM16/QAM64 and QPSK/8PSK pairs show significant confusion due to similar constellation structures — a known challenge in AMC literature.

Skills Demonstrated

- Deep learning implementation (PyTorch): CNN, LSTM, hybrid architectures
- Signal processing: I/Q data handling, SNR-stratified analysis
- Experimental methodology: Train/val/test splits, per-SNR evaluation, confusion analysis
- Data visualization: Training curves, accuracy plots, confusion matrices
- Technical documentation: Jupyter notebooks, version control (Git/GitHub)

Research & Industry Relevance

This project addresses challenges directly relevant to 6G wireless systems, cognitive radio, and spectrum sensing applications. Automatic modulation classification enables:

- Dynamic spectrum access and interference management
- Signal intelligence and spectrum monitoring
- Adaptive modulation and coding in next-generation networks

The methodology and findings transfer readily to other wireless signal classification tasks.

Repository & Resources

GitHub: github.com/nabeegh-khan/amc-deep-learning

Dataset: RadioML 2016.10a (DeepSig / GNU Radio)

Full Report: Available in repository docs/ folder